

ePaper: a look into the e-Future

By Vivian Nguyen



Contents

01

What is E-Paper

02

Applications

03

**E-Ink: Internal
Technology**

04

**Comparison to
LCD/LED**

05

**Future/Incoming
Innovations**

06

Q&A

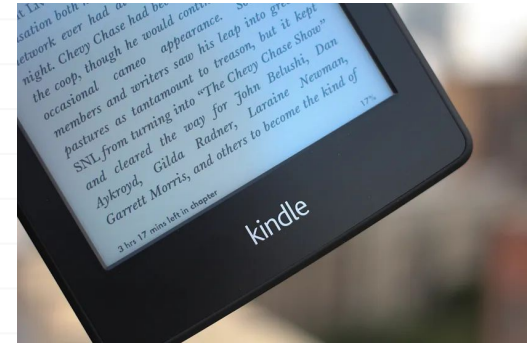
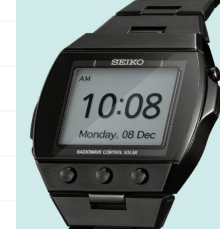
E-Paper Explained

- Electronic Paper
 - Electrophoretic Displays (EPDs)
- Reflective display technology
 - Does not emit light but rather reflects external light
- Designed to mimic the look of traditional printed ink on physical paper
- Sustainable alternative to paper
 - Reducing paper waste
 - Low energy usage



E-Paper Applications

- E-readers
 - Tablet made for reading digital books
 - monopolized by Amazon – Kindle
 - Thin tablet-like devices
- More commercially used for digital signage
 - Public transportation advertisements
 - Maps
 - Digital menus
 - Product price tags
- Smart-Watches, Clocks



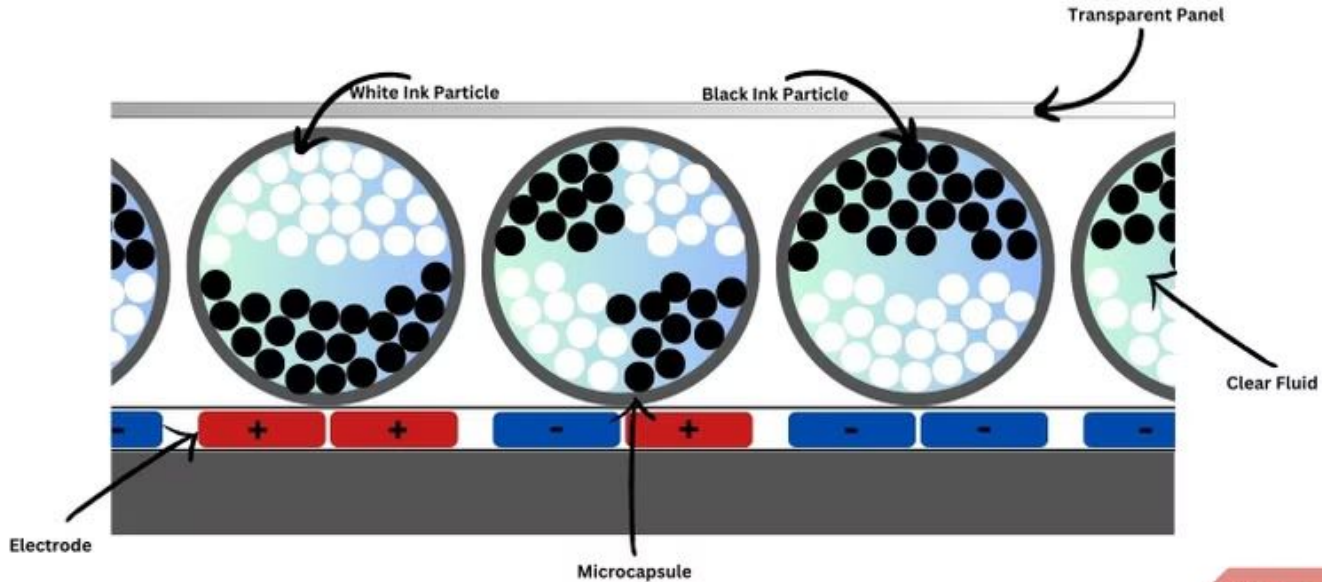
E-ink: the tech inside E-Paper



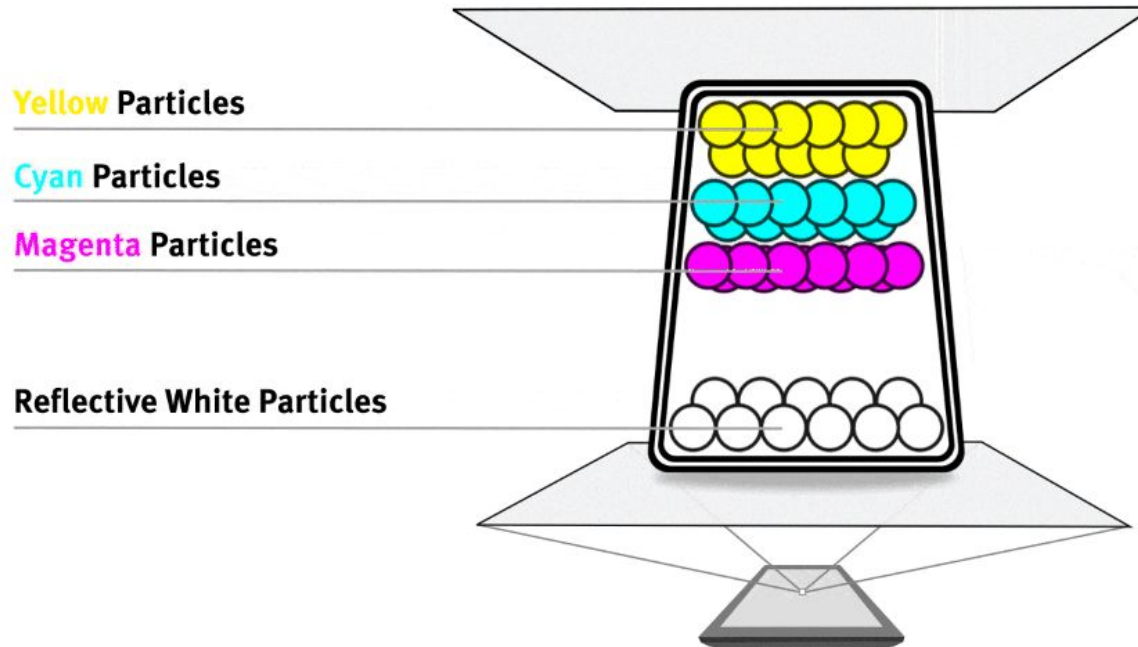
- E-Ink: Electrophoretic Ink or Electronic Ink
 - Technology is patented by E-Ink
- Colors and images that appear use real pigment
 - Millions of microcapsules of electronically charged pigment floating in a liquid substance
 - As thin as human hair
- Signal is sent → specific electrodes charged → image appears
 - EX. Positive electrode pulls down negatively charged pigment
 - Positively charged pigment rises to the surface
 - If pigment is dark:
 - capsule becomes a visible pixel on screen
- Power is only used to change image
 - Image is retained until the next action/update is made on the screen



E-Ink: Visual Explanation



E-Ink: Visual Explanation



What about LCD/LED Screens?

- LCD – Liquid Crystal Display
- LED – Light Emitting Diode
- Found in most general use screens
 - TV's, laptop screens, monitors, smartphones, tablets, etc.
- Large variety from different brands around the world



Compared to LCD/LED Screens

EPDs

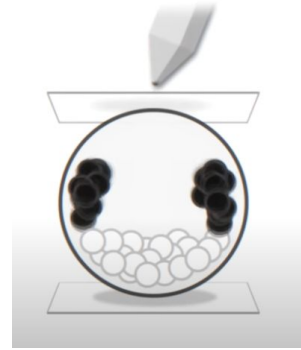
- Ideal for static, unchanging content
 - Text and images
- Dull colors
- Uses little energy consumption
 - Long battery life
- No glare, uses external lighting
 - No eye strain
 - High readability outdoors
- Noticeably weaker refresh rate
 - Flickering
- Economically friendly
- Difficult to troubleshoot/fix

LCD/LED

- Ideal for high-motion content
 - Videos, scrolling
- Bright color displays
- High energy consumption
- Glare, emits light
 - Eye strain
- Immediate refresh without flickering
- Easy to troubleshoot/fix

New/Upcoming Innovations

- E-Paper is coming back into relevance
- REMarkable
 - E-Paper tablet that allows you to write on it with a pressure sensitive pen
 - Streamlined writing/drawing usage
 - Immediate charge reactions
 - Cloud sync system to save notes/drawings
- More colored e-ink technology
 - Better image visibility
- Flexible Screens
 - Increasing adaptability and portability





Thank You



Questions?

Information/Image Resources

1. "How E-Paper Displays Works." *Pervasive Displays*, 12 Aug. 2024, www.pervasivedisplays.com/how-e-paper-works/. Accessed 01 Oct. 2024.
2. Pocketnow. "How E-Ink Works: The Technology Behind E-Paper Displays | Pocketnow." *YouTube*, 20 Jan. 2015, youtu.be/6NRT8E5-reM?feature=shared. Accessed 02 Oct. 2024.
3. TechAltar. "Why E-Ink Innovation Is so Slow." *YouTube*, 21 Dec. 2022, www.youtube.com/watch?v=8Xr9X6cbQ68. Accessed 02 Oct. 2024.
4. "Remarkable Paper Pro." *reMarkable*, remarkable.com/store/remarkable-paper/pro/details/features. Accessed 02 Oct. 2024.
5. Electronic Shelf Label | Retail | E Ink Applications. "Electronic Shelf Label." *E Ink Applications*, www.eink.com/application/detail/ESL. Accessed 02 Oct. 2024.
6. Supernote. "The History of Electronic Paper." *Medium*, 29 July 2019, medium.com/@supernote/the-history-of-electronic-paper-cdafa0d6cc38. Accessed 03 Oct. 2024.
7. "Displays - E-Paper Innovation Ltd." *E-PI*, 13 Aug. 2024, www.e-pi.com/displays/. Accessed 03 Oct. 2024.
8. Lum, Kelly. "Mainstream Digital Signage Vendors Tapping into E-Paper Displays for Sustainability Initiatives." *Omdia*, 4 June 2024, omdia.tech.informa.com/blogs/2024/jun/mainstream-digital-signage-vendors-tapping-into-e-paper-displays-for-sustainability-initiatives. Accessed 03 Oct. 2024.
9. "Electronic Shelf Labels." *Electronic Shelf Labels / Digital Price Tags*, 3 Sept. 2024, www.electronicshelftags.com/. Accessed 03 Oct. 2024.
10. Maning, Jayric. "LCD vs. Epaper: What's the Difference and What's Best for Reading?" *Make Use Of*, 11 Aug. 2023, www.makeuseof.com/lcd-vs-epaper-whats-best-for-reading/#:~:text=While%20LCDs%20are%20better%20for,value%20a%20better%20reading%20experience. Accessed 04 Oct. 2024.
11. Electronic Ink | E Ink Technology. *Electronic Ink | e Ink Technology*, www.eink.com/tech/detail/How_it_works#:~:text=Color%20is%20achieved%20by%20having,Gallery%20and%20Prism%20product%20lines. Accessed 04 Oct. 2024.
12. Colbrow, Brad. "The Kindle Scribe -VS- The ReMarkable II." *YouTube*, 27 Feb. 2023, www.youtube.com/watch?v=7guHaRIVzNU. Accessed 16 Oct. 2024.

